What publication delay costs: a pre-registered evaluation of day-ahead forecasting for India's State power systems

Rohit Kumar[a,1,*]

[a]*Centre for Artificial Intelligence, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, Jalandhar, 144008, Punjab, India*

Abstract

State control areas in India forecast demand, drawal, generation and prices from public data that arrive one to fourteen days after the event, and some sources drop out altogether. We treat this delay as part of the forecasting problem. In a State-day panel of 104,550 rows built only from official reports, every variable enters a forecast only after its measured publication delay. Running the same model as if all data were published at once shows that waiting costs between 15.1 % and 137.7 % of accuracy across five targets. We combine thirteen forecasters, among them pretrained models with covariates and a context model that weighs each source by how stale it is, and fix every choice in a pre-registration before a seventeen-month evaluation period is opened. Scored once on 34 State control areas, the final forecaster beats the seasonal naive and the single model chosen a year earlier on all five targets, and its 80 % and 90 % intervals stay within five points of nominal coverage.

*Corresponding author.

Email address: therohitkumar.build@gmail.com (Rohit Kumar)

[1]ORCID: https://orcid.org/0009-0004-7936-9683

Staleness-aware gating helps on three of five targets; three further context claims, including graceful degradation when a source stops, are not supported and are reported as findings. Of the 38 pre-registered checks reported here, 27 are supported. In an additional analysis that was not pre-registered, we pass every forecaster through a frozen day-ahead scheduling rule: each 0.1 GWh of mean absolute error per State-day costs about 0.26 Rs crore per day of system-wide settlement regret.
Keywords: day-ahead forecasting, probabilistic forecasting, publication delay, forecast combination, context gating, Indian power system

1. Introduction

Day-ahead decisions in India's power system are taken State by State. Each State control area schedules its drawal from the grid, procures power in the exchanges, and settles deviations from its schedule under national regulations [1, 2]. Its State Load Despatch Centre must estimate demand a day ahead and compute and publish its forecast errors, and the Grid Code leaves the forecasting method to each centre [3, Regulation 31]. Forecasts of demand, drawal, generation, renewable output and prices therefore matter at the State level, all the more as variable renewable generation grows in a coal-dominated system whose flexibility is limited [4, 5], and as State demand itself keeps rising [6]. The information needed to produce these forecasts is public. The grid operator, the Central Electricity Authority and the India Meteorological Department publish daily reports on demand, scheduling, station generation, renewable generation, prices and weather [7–11].

Public data, however, do not arrive when a forecasting benchmark as-

sumes they do. We measured the publication times of these sources:

- Power Supply Position reports appear about half a day after the data day, and later at weekends.
- Station generation arrives the next evening.
- The files carrying exchange prices and deviation rates arrive in weekly batches, a week or more after the date.
- Sources also drop out. In November 2025 the Central Electricity Authority moved its State renewable generation report to a new portal, so the listing that our pipeline read carried no report after 18 November; gridded weather ends with the last published year, and daily generation reports are missing for the 2020 lockdown months.

A model that treats every input as available up to the forecast origin uses information that no utility, regulator or researcher had at that time. A model that relies on a context source must also cope when that source stops.

This paper studies State-day forecasting under these real information sets, together with calibrated uncertainty and context fusion. It uses a provenance-tracked panel built only from official reports, which is openly available with the code [12] and documented in a companion data descriptor. A companion paper uses the forecasts studied here to schedule drawal a day ahead under the deviation settlement; this paper is about the forecasts themselves.

The contributions are:

1. **Measured information sets.** Each variable enters the forecast only after its measured publication delay. We quantify the cost of publication latency by rerunning the same model as if all sources published instantly.

2. **Latency-aware multi-context adaptive gating (LA-MCAG).** A forecaster in which each context block has its own encoder and a gate that sees how stale that source is. It is trained with random source dropout, so that it keeps forecasting when a source stops; whether it degrades more gracefully than matched fixed fusion is one of the pre-registered checks, and it was not supported. The gate is studied along an adaptivity ladder: fixed, per regime and per instance.

3. **Adaptation in time.** Two mechanisms use only residuals that have already been published: an online gate that rescales the context correction, and rolling conformal calibration. Both address a failure we observed in development: relationships learned over 2018–2023, such as demand driving prices, weakened after rapid solar growth.

4. **A pre-registered confirmatory evaluation** on 17 months of blinded data, with frozen model choices, multiplicity control and five seeds.

5. **What accuracy is worth.** In an additional analysis, kept apart from the pre-registered tests, every forecaster is passed through a frozen day-ahead scheduling rule, which puts a settlement-cost figure on forecast accuracy.

2. Related work

Energy forecasting. Load, generation and price forecasting are mature fields with their own reviews and benchmarks [13–15]. Probabilistic forecasting has been tested at scale in the Global Energy Forecasting Competitions [16], and open benchmarks for day-ahead prices show how much the conclusions of a study depend on its data and evaluation period [17]. Gradient-boosted

trees [18] and neural sequence models remain strong: recurrent networks [19], the Temporal Fusion Transformer, which combines variable selection with multi-horizon quantile outputs [20], patch-based transformers [21] and basis-expansion networks, which forecast the monthly demand of European countries better than statistical and hybrid benchmarks [22]. Pretrained foundation models such as Chronos provide zero-shot probabilistic forecasts [23], and large time series models have been fine-tuned for load forecasting where history is scarce [24]. A forecasting system adopted by the Portuguese distribution operator shows how far interpretable models with calendar and weather variables and a simple ensemble go, from system level down to secondary substations [25]. For India, deep learning has been applied to the demand of a single Union Territory [26], and State-wise medium- and long-term demand has been projected with weather, cooling and electric-vehicle drivers [6]; day-ahead forecasting for all State control areas from the official daily reports has, to our knowledge, not been studied.

Data that arrive late. That the latest values of some inputs are missing at the forecast origin is a central problem of macroeconomic nowcasting, where it is known as the ragged edge of the data set [27, 28]. There the release calendar of every series is part of the model. Energy forecasting studies rarely do the same: benchmarks almost always assume that every input is available at the forecast origin and that input sources persist through the test period. A recent exception uses publication timing in the other direction: prices of neighbouring European bidding zones are published before the local gate closure, and using them improves day-ahead price forecasts [29]. Methods for multivariate series with missing values [30] and imputation pipelines for in-

complete weather-station data [31] handle scattered gaps, not the systematic delay of a whole source at every forecast origin, which is what public power data in India have.

Combining forecasts. Combining forecasts is one of the most reliable ways to improve accuracy [32, 33], as large competitions keep confirming [34]. Quantile forecasts can be combined level by level [35, 36] or with weights fitted to a probabilistic score [37, 38], and online aggregation of experts updates the weights as losses arrive [39, 40], which helps load forecasts through distribution shifts [41]. When the losses themselves are published late, the feedback on which the weights depend is delayed as well, which the combination rules of this paper take into account.

Uncertainty and evaluation. Quantile forecasts are scored with proper scoring rules [42]. Conformalised quantile regression gives finite-sample coverage under exchangeability [43], adaptive conformal inference tracks coverage under distribution shift [44], and conformal methods have been carried into day-ahead regional wind power [45] and into electricity price forecasting with on-line recalibration [46]. Point accuracy is scaled against a seasonal naive [47], and forecast comparisons use the Diebold–Mariano test [48] with the small-sample correction of Harvey et al. [49]; good practice in forecast evaluation is reviewed by Petropoulos et al. [50].

The value of accuracy. Improvements in forecast accuracy matter to a power system only through the decisions they support. For wind power the value of better forecasts has been measured in system operation costs [51], and for a solar plant in the result of an optimal bidding strategy across day-

Table 1: Lines of work and what this paper adds.

Line of work	What it provides	What this paper adds
Energy forecasting models and benchmarks [13, 17, 22, 24]	Strong model families; open benchmarks	Every input enters with its measured publication delay; all 34 Indian State control areas, from official reports
Ragged-edge nowcasting [27–29]	Release calendars inside the model	Measured delays of public power data; sources that stop for good
Forecast combination [36, 40, 41]	Level-wise and online weights	Weights updated only from losses that are already published
Conformal calibration [43, 44, 46]	Coverage under exchangeability or under drift	Rolling calibration on published residuals, checked on blinded months
Value of forecasts [51, 52]	Cost or revenue of accuracy	Settlement regret of State drawal schedules under the Indian deviation settlement

ahead, intraday and imbalance markets [52]. We measure it for State drawal forecasts, through the deviation settlement that a State actually pays.

Gap. Table 1 places the paper among these lines of work. We are not aware of a study that measures the publication delays of a national power system's public data, builds them into the information set of every forecast, designs context fusion to be robust to source staleness and loss, tests all of this on blinded data under a pre-registration, and prices the resulting accuracy in the settlement.

3. Problem setting

Targets. For State control area n and day t we forecast:

T1 energy met;

T2 actual drawal, which is the input to scheduling;

T3 conventional generation;

T4 renewable generation, where published;

T5 the day-ahead market clearing price of each exchange bid area.

Information sets. A forecast is issued at noon on day τ for days $\tau+1$, $\tau+2$ and $\tau+3$ (horizon $H = 1, 2, 3$). Each variable x has a measured availability delay L_x in days.

Definition 1 (Published information set)**.** *The information set at issue day τ is $\mathcal{I}_\tau = \{x_d : d \leq \tau - L_x\}$ over all variables x and dates d, together with calendar information about the target days.*

With $L = 1$ for grid operator and weather variables, $L = 2$ for generation and renewable variables and $L = 14$ for price files, a price forecast for $\tau + 1$ is effectively a 15-day-ahead forecast. Instant publication ($L_x = 0$ for all x) defines the counterfactual used to measure the cost of latency.

Staleness. For a context block k with key variable x_k, the staleness at τ is the number of days since x_k was last observed within $\mathcal{I}_\tau$. It is zero-to-small when a source publishes normally and grows without bound when a source stops.

Adaptivity. Write the forecast as a base forecast from the target's own history plus a context correction, $\hat{y} = b + g \cdot r$. The gate g can be fixed, differ by regime, differ by instance (a function of the inputs), or change over time. A finer gate reduces bias when the optimal weight truly varies, and adds estimation variance when it does not. Whether a finer gate pays, and whether

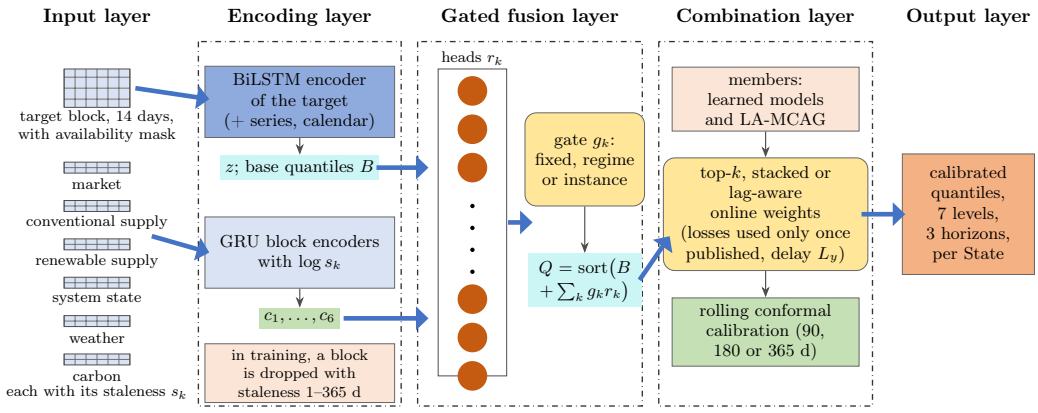


Figure 1: Forecasting pipeline. Each input block enters with its own publication delay; the target and the six context blocks are encoded separately, each block's correction of the base quantiles is scaled by a gate (fixed, by regime, or per instance), and the forecasts of all members are combined with weights that use only losses already published, then calibrated by rolling conformal shifts.

that can be predicted from calibration data alone, is the subject of a companion study; here the gates are compared only through the accuracy they deliver.

4. Method

Figure 1 shows the forecasting pipeline: the gated architecture (LA-MCAG) and the combination of forecasters that follows it.

4.1. LA-MCAG architecture

Inputs are 14-day windows of the published information set, aligned so that each variable enters with its own delay. The target block holds the target and its availability mask. The context blocks are:

- market: deviation rate and exchange prices;

- conventional supply;
- renewable supply;
- system state and reliability: demand, schedules, deviations, frequency stress, outages, coal stock;
- environment: weather;
- carbon: carbon intensity of State generation, the carbon layer.

Each window is normalised at window level. The target is centred on the mean of its last seven available values and scaled by a fixed series scale. Each context column is split into a within-window shape and a window level. This lets the model follow level shifts, such as growing renewable capacity, instead of extrapolating training means.

A bidirectional LSTM encodes the target block into a representation z, together with a series embedding and the calendar of the target days. A linear base head gives base quantiles B. Block k is encoded by a GRU and concatenated with its log-staleness s_k to give c_k. A correction head gives $r_k = f_k(z, c_k)$. The output quantiles are

$$Q = \operatorname{sort}\Big(B + \sum_k g_k\, r_k\Big), \qquad g_k \in (0, 1.5). \tag{1}$$

Sorting the seven levels is the monotone rearrangement of Chernozhukov et al. [53]. It removes crossed quantiles and never increases the loss. For an outcome y and levels $\alpha_1 < \cdots < \alpha_J$, with $\rho_\alpha(u) = u\,(\alpha - \mathbf{1}\{u < 0\})$, the total pinball loss is

$$\sum_{j=1}^{J} \rho_{\alpha_j}(y - q_j) = \sum_{j=1}^{J} \alpha_j\,(y - q_j) + \sum_{j=1}^{J} (q_j - y)^+. \tag{2}$$

The second sum does not depend on the order of the q_j, and by the rearrangement inequality the first is smallest when the q_j are in increasing order,

so the sorted quantiles score at least as well as the unsorted ones on every day. The same holds for every later step that sorts.

The gate takes one of three forms:

- **fixed:** $g_k = 1.5\,\sigma(\theta_k)$;
- **regime:** $g_k = 1.5\,\sigma(\theta_{k,\rho})$, where ρ indexes season and an ex-ante price-volatility state;
- **instance:** $g_k = 1.5\,\sigma(h_k(z, c_k, s_k))$, the multi-context adaptive gate made staleness-aware.

The comparator *fixed fusion* concatenates all c_k into a single head, with hidden width chosen so that its parameter count matches the instance-gated model.

During training, each context block is replaced by "missing" with probability 0.15: values zero and masks one. In the first development round a dropped block always carried a staleness of 30 days. A source that stops for good, however, grows stale for months, which the model then never saw in training. The second round therefore draws the staleness of a dropped block from a log-uniform distribution between 1 and 365 days (variants marked *sd*); the matched fusion model is trained the same way. Training minimises the pinball loss over seven quantiles and three horizons.

4.2. Adapting in time with published residuals

Online time-adaptive gate (OTG). Let $R = y - b_{0.5}$ and $r = q_{0.5} - b_{0.5}$ be the residual and the context correction at the median. For series n and horizon H, the gate at issue day τ is

$$\hat{g}(\tau) = \mathrm{clip}_{[0,1.5]}\left(\frac{\sum_{d\in W_\tau} R_d\, r_d}{\sum_{d\in W_\tau} r_d^2}\right), \qquad W_\tau = \{d : \tau - L_y - 180 < d \le \tau - L_y\}, \tag{3}$$

so that W_τ holds only target dates whose outcomes are already published at τ. The ratio is the least-squares slope of R on r through the origin, $\arg\min_g \sum_{d \in W_\tau} (R_d - g\, r_d)^2$: $\hat{g} < 1$ shrinks a context correction that has recently overshot and $\hat{g} > 1$ amplifies one that has fallen short. If fewer than 30 residuals are available, or $\sum r_d^2 = 0$, then $\hat{g} = 1$. The released quantiles are $\mathrm{sort}\big(B + \hat{g}\,(Q - B)\big)$; the sort matters when $\hat{g} > 1$ or when the base quantiles B cross.

Rolling conformal calibration. For each quantile level α, the shift $c_\alpha(\tau)$ is a conformal quantile of the residuals $e_d = (y_d - q_{\alpha,d})/s$, scaled by the series scale s, over the same kind of published window, $\tau - L_y - w < d \le \tau - L_y$ with w of 90, 180 or 365 days. With the m residuals in increasing order, c_α is the $\lceil (m+1)\alpha \rceil$-th for $\alpha \ge 0.5$ and the $\lfloor (m+1)\alpha \rfloor$-th for $\alpha < 0.5$, the finite-sample ranks of split conformal prediction [43]. The released quantile is $q_\alpha + s\, c_\alpha$, and the levels are sorted again. Shifts are computed per series and horizon; a series with fewer than 30 residuals in the window uses the residuals of all series at that horizon. Models without quantile heads are calibrated around their median in the same way. Because the window rolls, coverage is not guaranteed under drift; it is checked on the blinded period instead (Section 7).

4.3. Combining forecasters

No single model family was best for every target in the first development round, and the price target (T5) changed behaviour when the 2024 deviation-settlement rules took effect. The second round therefore combines forecasters. Members are the learned models of Section 5 and the better

LA-MCAG variant. Their quantiles are combined level by level, which keeps the combined quantiles in order [35]; members without the 5% and 95% levels are extended linearly from their outer quantiles. Three combinations are formed.

- **Top-k average:** equal weights on the k members with the lowest calibration-year error.
- **Stacked weights:** one non-negative weight per member and horizon, summing to one, fitted by minimising the pinball loss on the calibration year with a penalty $\lambda\|w - \frac{1}{M}\mathbf{1}\|^2$ towards equal weights, in the spirit of quantile regression averaging [36].
- **Lag-aware online weights:** at issue day τ the weight of member m is

$$w_m(\tau) \propto \exp\Big(- \eta \sum_{d \le \tau - L_y} \gamma^{\,(\tau - L_y) - d}\, \bar{\ell}_m(d) \Big), \tag{4}$$

  where $\bar{\ell}_m(d)$ is the mean scaled pinball loss of member m on target date d, pooled over all series or over the series of a region, L_y is the publication delay of the target and $\gamma \le 1$ a forgetting factor. Only losses that are already published enter the sum. This is the exponentially weighted average forecaster of prediction with expert advice [39, 54], with discounting of old losses as in online aggregation under drift [40, 41] and the one change that matters for public data: the feedback arrives L_y days late. Weights are normalised to sum to one, recomputed for every issue day and horizon, and the weighted quantiles are sorted.

The settings k, λ, η, γ and the pooling are chosen on the calibration year by two-fold cross-validation over alternating ISO weeks, and so is the final forecaster of each target: the best single member, top-k, stacking or online

weights, whichever has the lowest cross-fitted calibration-year MASE. Cross-fitting keeps the in-sample choices (best member, top-k set, stacked weights) from looking better than the online weights, which are scored as they run. The combined quantiles are then calibrated by rolling conformal shifts as above, with the window (90, 180 or 365 days) chosen on the calibration year.

5. Data and experimental design

Data. The State-day panel covers 34 State control areas (all 28 States, five Union-Territory control areas and the Damodar Valley Corporation) from April 2018 to August 2026 and is built only from official reports of Grid-India, the Central Electricity Authority and the India Meteorological Department. It is openly available with the code [12] and documented in a companion data descriptor. Every variable carries its measured availability delay.

Models.

- **Anchors:** persistence, same-weekday seasonal naive and a seven-day mean, all built from the latest available values.
- **Gradient-boosted trees:** LightGBM direct quantile models per horizon [18] and XGBoost multi-quantile models [55], on the same lag-aware features.
- **Recurrent and attention models:** a bidirectional LSTM quantile model, TFT [20] and PatchTST [21].
- **Multilayer and convolutional models with covariates:** N-HiTS [56], TiDE [57], BiTCN [58] and N-BEATSx [59], as implemented in

`neuralforecast`, each with the lag-aligned past covariates and the calendar as known future inputs.

- **Pretrained models, zero-shot:** Chronos-Bolt Small and Base [23] on the target history, and Chronos-2 [60] with the same past covariates and the calendar.
- **LA-MCAG variants** and the combinations of Section 4.

Tuning grids and seeds were fixed before fitting. Trained models use five seeds, averaged quantile-wise; the input windows of the covariate models (28 or 56 days) and the tree depth of XGBoost (6 or 8) were chosen on the calibration year with seed 0.

Development and confirmation. Development ran in two rounds, both logged with their reasons before results were read. Models were fitted on FY2018-19 to FY2022-23 (the last 180 days used only for early stopping), choices were made on FY2023-24, and results were read on FY2024-25. The second round added the models and combinations above and the staleness-matched training; because FY2024-25 had been read once already, every second-round choice was made on FY2023-24 only.

The confirmatory evaluation repeats the pipeline with every boundary moved forward one year:

- fit on FY2018-19 to FY2023-24;
- calibration, combination weights and online-gate warm-up on FY2024-25;
- evaluation on 1 April 2025 to 31 August 2026.

The evaluation data stayed behind a blinding guard in the code until the pre-registration was frozen: its SHA-256 hash, together with the hashes of every frozen choice file and of the scoring script, was committed with a git

tag before the first evaluation row was loaded. All choices are frozen from development; procedures that fit numbers on the calibration year (combination weights, conformal shifts) keep their frozen settings. Deviations after the freeze are listed in a dated log. A leakage test is one stage of the pipeline: for sampled targets, States and issue days τ, every panel value dated after $\tau - L_x$ is replaced by noise, and the features and anchors of issue day τ must stay identical; the test passed for all five targets in the confirmatory run.

Metrics and tests.

- **Point accuracy:** mean absolute scaled error against the seasonal naive at the same lead [47].
- **Probabilistic accuracy:** pinball loss and empirical coverage of 80% and 90% intervals [42].
- **Comparisons:** daily cross-series means of scaled loss differentials, tested with HLN-corrected Diebold–Mariano statistics [48, 49] and seven-day block-bootstrap confidence intervals [61].
- **Multiplicity:** Holm within each family of hypotheses [62] and Benjamini–Yekutieli across families [63].

Computation. Every stage writes a record of its duration, and the code at the freeze is tagged in the project repository. The forecasting stages of the confirmatory run took 4.8 hours in all; Table A.5 gives the time by model group.

6. What forecast accuracy is worth

A State uses its drawal forecast to fix a schedule, so the accuracy of that forecast has a price in the settlement. To put a figure on it we take the

day-ahead scheduling rule of the companion paper and do not change it. For State n on day t the rule chooses the schedule S that minimises the expected settlement regret over the calibrated forecast distribution of actual drawal X, widened by a factor $1+\rho$ around its median; ρ and the risk weight were fixed in development. In a stylised form of the 2024 deviation settlement [2] the regret of a schedule is

$$\ell(S,X) = p\,(S-X) + \lambda^{+}\big[\min(D^{+},b) + 1.2(D^{+}-b)^{+}\big] \\ - \lambda^{-}\big[0.9\min(D^{-},b) + 0.8(D^{-}-b)^{+}\big] \;\geq\; 0, \quad (5)$$

where p is the day-ahead price, λ the deviation rate, $\lambda^{+}=\max(\lambda,p)$, $\lambda^{-}=\min(\lambda,p)$, $D^{+}=(X-S)^{+}$ and $D^{-}=(S-X)^{+}$ the over- and under-drawal, and $b=0.1|S|$ the tolerance band. The regret is the settlement cost of the schedule minus the cost pX of buying the actual drawal day-ahead, so a perfect schedule has $\ell(X,X)=0$. It cannot be negative. When $X>S$, the first bracket is at least D^{+} and $\lambda^{+}\geq p$, so $\ell \geq -pD^{+}+pD^{+}=0$. When $X<S$, the second bracket is at most $0.9\,D^{-}$ and $0\leq\lambda^{-}\leq p$, so $\ell \geq pD^{-}-pD^{-}=0$ (prices and deviation rates are non-negative). The rule sees only prices published before the decision, and every schedule is kept inside the State's historical operating envelope. The system-wide regret of a day is the sum over States.

We pass the day-ahead drawal forecast of every forecaster of Section 4, the thirteen members and the three combinations, each calibrated on a rolling 180-day window, through this rule on the evaluation period. For each we record the mean absolute error of the forecast median and the mean system-wide regret of the schedules, and we fit the regret against the error across

forecasters; the slope is re-estimated on seven-day block resamples of the evaluation days. As a check, the final forecaster run with its own calibration window must reproduce the confirmatory result of the companion paper; it does, for every rule, and the analysis would stop if it did not.

This analysis was designed after the confirmatory run and is not part of the pre-registration. It fits no new model and tunes nothing on the evaluation period, but it is descriptive: the forecasters differ in more than their error, so the slope is an association across forecasters, not the effect of an intervention. Its script (`codes/scripts/extra/x01_value_of_forecast.py`) and outputs (`results/extra/value_of_forecast`) are in the public code repository named in the code availability statement; they postdate version 1.0 of the Zenodo record.

7. Results

All numbers in this section come from the single confirmatory run on 1 April 2025 to 31 August 2026, scored once under the frozen pre-registration; development numbers, where they are quoted, are labelled. The pre-registration holds 52 checks in five families, of which 35 are supported. This paper reports the two families about forecasting: forecasting itself (20 of 20 supported), context gating and the carbon layer (7 of 18), 27 of 38 in all. The price-of-adaptivity family (1 of 3) is reported in a companion study of gated forecast correction. The scheduling family (6 of 9) and the deviation-assessment family (1 of 2), whose main test failed, are reported in the companion scheduling paper. Appendix Appendix A lists every check of the two families reported here.

7.1. Forecasting (F1): supported for every target

Table 2 and Fig. 2 give the accuracy of the final forecaster of each target against the seasonal naive and against the model selected in the first development round. Every target improves on both, and every one of the five seeds is better than the seasonal naive.

Table 2: Confirmatory accuracy. MASE uses the training-period seasonal-naive scale, so the seasonal naive itself is not 1 on this period. Coverage is of the calibrated 80 % and 90 % intervals.

Target	Final forecaster	MASE	First round	Seasonal naive	Cov. 80 %	Cov. 90 %
T1 energy met	ens-stack	0.892	0.927	1.338	0.789	0.893
T2 actual drawal	ens-stack	0.922	0.953	1.339	0.792	0.893
T3 conventional generation	ens-top	0.859	0.862	1.130	0.802	0.899
T4 renewable generation	chronos2	1.107	1.164	1.615	0.778	0.886
T5 day-ahead price	ens-top	0.741	2.598	1.027	0.786	0.892

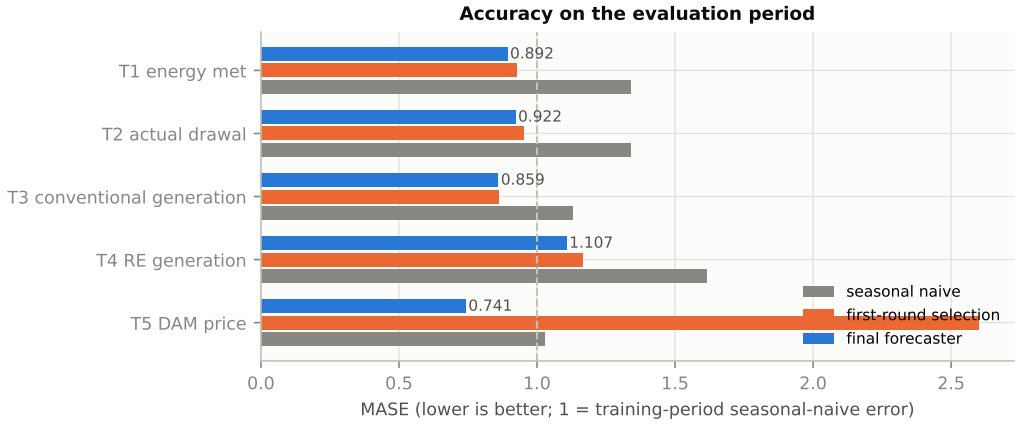


Figure 2: Accuracy on the confirmatory period against the seasonal naive and the first-round selection.

The gain over the first-round model is largest for the price target, where the first-round choice (a BiLSTM picked on a calibration year that did not

contain the new settlement rules) reached MASE 2.598 while the combination reached 0.741 (paired difference -1.8576 in scaled absolute error, Holm-adjusted $p = 1 \times 10^{-95}$). The same pattern, much smaller in size, holds for the other four targets (H1f supported throughout; for energy met the difference is -0.0346, $p = 3 \times 10^{-20}$). Intervals stay inside the pre-registered five-point band at both levels for all five targets (H2).

Table 3 gives every member on every target. Two points stand out. Chronos-2 with covariates is the best single model on all five targets, ahead of the tuned gradient-boosted trees. And the frozen selection rule did not always pick the combination that turned out best: for conventional generation and for the price it chose the top-k average, while stacked weights and the lag-aware online weights were lower on the evaluation period. This is the cost of making every choice on the calibration year only, and we report it as it is. The supplementary material completes this picture: the pinball loss of every quantile forecaster (Table S1), every pairwise forecast comparison with its test (Table S2), the cross-fitted scores behind the choice of the combination (Table S3), all gated and fused context variants and their accuracy when sources are lost (Tables S4 and S5), and the spread of accuracy between State control areas (Table S6).

7.2. The cost of late public data (F1, H3)

Waiting for official publication is expensive. With measured delays the same LightGBM model loses 16.2 % accuracy on energy met, 15.1 % on actual drawal, 30.0 % on conventional generation, 15.8 % on renewable generation and 137.7 % on the day-ahead price, compared with the same model given the data immediately (Fig. 3); the smallest cost is 15.1 % (actual drawal) and

Table 3: Point accuracy of every forecaster on the confirmatory period (MASE against the training-period seasonal-naive scale, pooled over horizons). Bold marks the forecaster that the frozen selection rule picked for that target.

Forecaster	T1 energy met	T2 actual drawal	T3 conventional generation	T4 RE generation	T5 DAM price
seasonal naive	1.338	1.339	1.130	1.615	1.027
LightGBM	0.927	0.953	0.862	1.291	1.123
XGBoost	0.932	0.954	0.860	1.309	1.293
BiLSTM	1.028	1.020	1.008	1.361	2.598
TFT	0.949	0.979	0.888	1.216	1.012
PatchTST	1.021	1.027	0.937	1.438	0.970
N-HiTS	1.014	1.016	2.040	1.277	1.041
TiDE	0.954	0.990	0.891	1.342	0.836
BiTCN	1.002	1.023	1.149	1.304	0.818
N-BEATSx	1.005	1.015	1.161	1.266	1.148
Chronos-Bolt Small	0.969	0.983	0.908	1.164	0.877
Chronos-Bolt Base	0.958	0.977	0.906	1.165	0.884
Chronos-2	0.918	0.941	0.853	**1.107**	0.711
LA-MCAG instance, staleness-matched	0.936	n/a	n/a	n/a	1.406
top-k average	0.901	0.925	**0.859**	1.141	**0.741**
stacked weights	**0.892**	**0.922**	0.831	1.131	0.717
lag-aware online weights	0.901	0.927	0.836	1.115	0.711
LA-MCAG (instance gate)	n/a	0.967	0.865	1.222	n/a

the largest 137.7 % (DAM price), where the price files arrive about a fortnight late and a day-ahead forecast is in effect a fifteen-day-ahead forecast. H3 is supported for every target. This is the quantitative case for treating publication delay as a first-class part of the forecasting problem rather than an implementation detail.

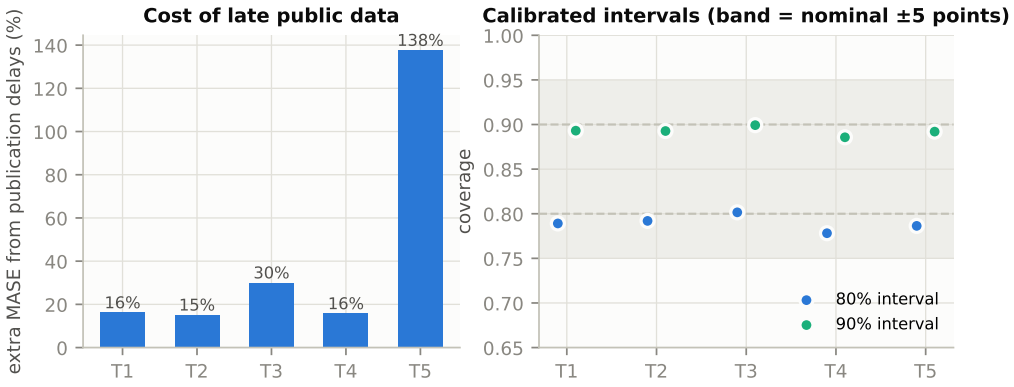


Figure 3: Left: accuracy lost to publication delays. Right: coverage of the calibrated intervals, with the pre-registered band.

7.3. Context gating and the carbon layer (F2)

Figure 4 shows the staleness-aware gated model against matched fixed fusion. On pinball loss the gated model is better for actual drawal (-0.0014, $p_{\text{Holm}} = 0.018$), conventional generation and the day-ahead price, and not better for energy met and renewable generation: H4 is supported for three of five targets. The remaining context tests are weaker. Real context beats null and shuffled context only for actual drawal (H5); the online time-adaptive gate improves point accuracy only for the price (H6). The source-loss test is not supported on either target, and the two targets fail it in different ways:

when the renewable and weather blocks are removed for the whole period, the gated model degrades slightly *less* than fusion on energy met (-0.0027, 95 % interval [-0.0055, -0.0002]), which would have passed on its own but not after the Holm correction inside the family ($p_{\text{Holm}} = 0.055$), while on actual drawal there is no sign of an advantage at all ($+0.0019$, $p_{\text{Holm}} = 1.000$). The pre-registered robustness claim is therefore not established, and whatever effect exists is small. The carbon block does no harm on energy met and actual drawal (H8 supported as non-inferiority) but its expected improvement on conventional generation is not confirmed. Context earns its place in this system through the gated model on three targets, not as a blanket claim.

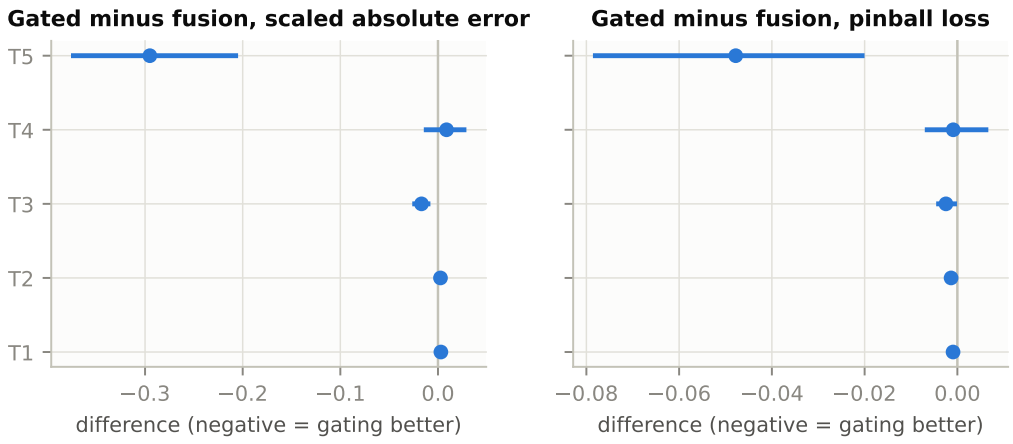


Figure 4: Gated context against matched fixed fusion, confirmatory period.

7.4. What accuracy is worth (additional analysis, not pre-registered)

Figure 5 shows, for each of the 16 forecasters of Section 6, the mean absolute error of its day-ahead drawal forecast and the mean system-wide regret of the schedules the frozen rule builds from it, over 516 evaluation

days. The relation is close to a straight line: the rank correlation between error and regret is 0.97, and each gigawatt-hour of mean absolute error per State-day costs 2.63 Rs crore per day of system-wide regret (95 % block-bootstrap interval [2.32, 2.89]), or about 0.26 Rs crore per day for each 0.1 GWh. The most accurate forecaster, the stacked combination (5.70 GWh), and the least accurate, BiTCN (6.56 GWh), are 2.23 Rs crore per day apart. The final forecaster, with its own calibration window, reaches 5.68 GWh and 15.80 Rs crore per day. The figure is a statement about this settlement and these forecasters, not a causal price of accuracy, but it gives the size of the stake.

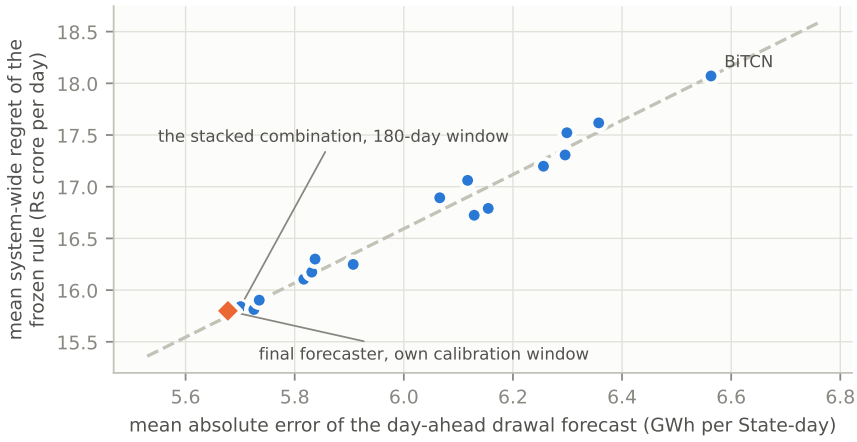


Figure 5: Accuracy of each day-ahead drawal forecaster against the settlement regret of the schedules a frozen rule builds from it, evaluation period. The dashed line is the least-squares fit across forecasters. Additional analysis, not pre-registered.

7.5. Where the results come from

The main numbers are averages over States, days and horizons, so we also show how they vary. Table 4 gives the accuracy gain over the seasonal

naive by season: the gain is present in every season of every target, largest in the post-monsoon months and smallest in winter, and renewable generation has no winter figure because that report left the source listing used here in November 2025. The table is descriptive: it was produced after the confirmatory scoring and is not a pre-registered test.

Table 4: Accuracy gain over the seasonal naive by season on the confirmatory period (per cent reduction in scaled absolute error). Descriptive; not a pre-registered test.

Target	Monsoon	Post	Summer	Winter
T1 energy met	32.4	40.2	33.1	29.6
T2 actual drawal	30.2	38.8	31.0	26.5
T3 conventional generation	24.4	28.5	23.4	19.7
T4 RE generation	27.6	32.8	36.5	–
T5 DAM price	28.8	22.9	28.2	26.1

8. Discussion

What the evidence supports. Two things hold on the blinded period. First, day-ahead State forecasts built only from official Indian data beat the seasonal naive on all five targets, with calibrated intervals; a combination of a pretrained model with covariates, gradient-boosted trees and the gated context model is better than any single first-round model (H1, H1f, H2). Second, the delay with which public data appear is a large and measurable cost, from 15.1 % to 137.7 % of accuracy (H3), which justifies designing around publication delay rather than assuming that data arrive on time.

What the evidence does not support. Several claims failed their pre-registered tests and stay in the paper as findings. Gating helps on three targets of five, not on all. Real context beats null and shuffled context on one target of three (H5), and the online time-adaptive gate helps on one target of five (H6). The claim that a gated model degrades more gracefully when sources disappear is not established (H7): on energy met the gated model loses slightly less than fusion, by an amount whose unadjusted test would have passed but which does not survive the Holm correction inside its family, and on actual drawal there is no advantage at all. The carbon block does no harm but does not deliver the improvement expected on conventional generation (H8).

Why the negative results are informative. The failures share a pattern: they are the parts that try to adapt a second time on top of an already adaptive system. Once the forecaster is a calibrated combination, the room left for context-dependent adjustments is small, and estimating them costs more than it returns. The same logic explains why the robustness claim is so hard to demonstrate: with staleness-matched training both models already handle a missing block, so the difference between them is a second-order effect that seventeen months of daily data cannot resolve.

What accuracy is worth. The additional analysis of Section 6 turns these accuracy differences into money. With the scheduling rule held fixed, schedules built from a less accurate forecast cost more in the settlement, almost in proportion to the error, at about 0.26 Rs crore per day for each 0.1 GWh of mean absolute error per State-day. The figure depends on a stylised settlement and on the particular set of forecasters compared, so it is an order of magnitude, not a tariff. It still makes the practical case plain: for a State load dispatch

centre, time spent on data timeliness and on forecast combination is spent where the settlement pays for it.

Limitations. The panel is daily; block-level (15-minute) data would allow intra-day forecasts that we do not study. Renewable generation data end on 18 November 2025, when the report moved to a new portal that the frozen pipeline did not read (the extended panel of the data record's next version adds the later reports), and weather ends on 31 December 2025, so the last months of the evaluation rely on the models' own handling of missing sources, and the renewable target has no winter figure. The second development round read the same development year as the first; all claims here rest on the blinded period, and the development numbers are reported separately. The frozen selection rule was applied without exception, which on two targets left a better combination unused; we report this and did not replace the frozen choice with the better one. The value of accuracy rests on a stylised settlement, with no frequency linkage, no 15-minute blocks and no volume limits, and on an association across forecasters.

Practical implications. For an Indian State load dispatch centre, the recipe that the evidence supports is: build the information set with the delays that the data really have, combine a small set of forecasters with weights set only from data that have already been published, calibrate the intervals on a rolling window, and use context gating where it has been shown to help rather than everywhere. The combination weights, the online gate and the calibration need only past forecast errors, which State Load Despatch Centres already compute and publish under the Grid Code [3, Regulation 31], and every input comes from reports that Grid-India, the Central Electricity

Authority and the India Meteorological Department publish, so a centre, a Regional Load Despatch Centre or the regulator can rerun the evaluation on its own data. The daily resolution is the limit: the Grid Code asks for 15-minute day-ahead estimates, for which the same information-set discipline would have to be repeated on block-level data.

Reproducibility. Every number here is produced by one driver script, recorded stage by stage, and written into the text as a macro from the result files. The pre-registration, the frozen choices and the scoring code were hashed and tagged before the evaluation data were opened. Re-running the scoring script on the stored results after the fact reproduces the hypothesis table cell for cell, and the panel still matches the content hash recorded when it was built. The additional analysis has its own scripts, reads only stored results, and writes its numbers to a separate file.

9. Conclusion

We set out to test, rather than assert, how well Indian State power systems can be forecast from the public data that actually exist: data that arrive late, from sources that sometimes stop. The test was pre-registered, run once on 518 days of data that had been kept closed, and scored by a script whose hash was fixed beforehand. Of the 38 checks reported here, 27 are supported, including all 20 in the forecasting family.

Three results are firm enough to act on. Publication delay has to be treated as part of the forecasting problem: reproducing the real delays costs between 15.1 % and 137.7 % of accuracy, more than the spread between most model families. A combination of a small set of forecasters, with weights

set only from data that had already been published, is better than choosing one model a year in advance, and the gain is largest exactly where a regime change broke the earlier choice. And the intervals, calibrated on a rolling window, stay close to their nominal coverage through a period in which one source dropped out.

Several context claims did not survive their own test, and we report them as results. Gating by staleness helps on three targets of five, not on all; real context beats a shuffled one on only one target of three; and a gated model does not degrade clearly more gracefully than matched fixed fusion when sources disappear, since the small advantage on energy met does not survive the correction for multiple tests and there is none on actual drawal. The pattern is consistent: once the forecaster is a calibrated combination, a second layer of adaptation has little left to earn and costs estimation variance.

An additional analysis, outside the pre-registration, puts a price on what the confirmed results deliver. Passed through a frozen day-ahead scheduling rule, every 0.1 GWh of mean absolute error per State-day costs about 0.26 Rs crore per day in the settlement. Block-level data and a prospective test on months that arrive after this study are the natural next steps.

Appendix A. Supporting tables

Tables A.1 and A.2 list every pre-registered check of the two families reported in this paper; the other tables support Sections 5 and 7.

Table A.1: Pre-registered checks of family F1 (forecasting), scored once on the confirmatory period (20 of 20 supported). Decision as pre-registered: supported when the Holm-adjusted p within the family is below 0.05 and the check's extra rule holds; Benjamini–Yekutieli across all checks is reported only.

ID	Statement	Estimate	95% CI	p_{Holm}	p_{BY}	Decision
H1a	T1: final forecaster beats seasonal naive (scaled |e|, H pooled)	-0.4448	[-0.5000, -0.3955]	1.8×10^{-86}	5.7×10^{-86}	supported
H1f-T1	T1: final forecaster beats the round-1 selected model (lgbm)	-0.0346	[-0.0429, -0.0258]	2.7×10^{-20}	7.3×10^{-20}	supported
H2-T1	T1: 80 % / 90 % coverage within ±5 points	+0.7891	[+0.7891, +0.8931]	–	–	supported
H3-T1	T1: publication-latency cost > 0 (LightGBM, % MASE)	+16.2044	[-0.1426, -0.1174]	4.3×10^{-106}	2.1×10^{-105}	supported
H1b	T2: final forecaster beats seasonal naive (scaled |e|, H pooled)	-0.4168	[-0.4680, -0.3721]	7.2×10^{-92}	2.5×10^{-91}	supported
H1f-T2	T2: final forecaster beats the round-1 selected model (lgbm)	-0.0318	[-0.0399, -0.0230]	5.9×10^{-19}	1.7×10^{-18}	supported
H2-T2	T2: 80 % / 90 % coverage within ±5 points	+0.7920	[+0.7920, +0.8927]	–	–	supported
H3-T2	T2: publication-latency cost > 0 (LightGBM, % MASE)	+15.1392	[-0.1369, -0.1142]	1.0×10^{-113}	7.2×10^{-113}	supported
H1c	T3: final forecaster beats seasonal naive (scaled |e|, H pooled)	-0.2701	[-0.3076, -0.2348]	1.3×10^{-61}	3.9×10^{-61}	supported
H1f-T3	T3: final forecaster beats the round-1 selected model (lgbm)	-0.0029	[-0.0052, -0.0009]	9.1×10^{-4}	0.008	supported
H2-T3	T3: 80 % / 90 % coverage within ±5 points	+0.8016	[+0.8016, +0.8993]	–	–	supported

ID	Statement	Estimate	95% CI	p_{Holm}	p_{BY}	Decision
H3-T3	T3: publication-latency cost > 0 (LightGBM, % MASE)	+30.0267	[-0.2165, -0.1830]	4.8×10^{-114}	6.2×10^{-113}	supported
H1d	T4: final forecaster beats seasonal naive (scaled |e|, H pooled)	-0.5080	[-0.7026, -0.3573]	5.9×10^{-7}	3.5×10^{-6}	supported
H1f-T4	T4: final forecaster beats the round-1 selected model (chronos)	-0.0574	[-0.0838, -0.0358]	1.5×10^{-9}	6.7×10^{-9}	supported
H2-T4	T4: 80 % / 90 % coverage within ±5 points	+0.7782	[+0.7782, +0.8857]	–	–	supported
H3-T4	T4: publication-latency cost > 0 (LightGBM, % MASE)	+15.7772	[-0.2249, -0.1331]	1.2×10^{-17}	3.8×10^{-17}	supported
H1e	T5: final forecaster beats seasonal naive (scaled |e|, H pooled)	-0.2860	[-0.3925, -0.1811]	5.1×10^{-12}	1.9×10^{-11}	supported
H1f-T5	T5: final forecaster beats the round-1 selected model (bilstm)	-1.8576	[-2.0822, -1.6347]	1.1×10^{-95}	4.4×10^{-95}	supported
H2-T5	T5: 80 % / 90 % coverage within ±5 points	+0.7864	[+0.7864, +0.8920]	–	–	supported
H3-T5	T5: publication-latency cost > 0 (LightGBM, % MASE)	+137.6633	[-0.8024, -0.5388]	1.4×10^{-43}	4.1×10^{-43}	supported

Table A.2: Pre-registered checks of family F2 (context gating and carbon), scored once on the confirmatory period (7 of 18 supported). Decision as pre-registered: supported when the Holm-adjusted p within the family is below 0.05 and the check's extra rule holds; Benjamini–Yekutieli across all checks is reported only.

ID	Statement	Estimate	95% CI	p_{Holm}	p_{BY}	Decision
H4-T1	T1: gated (sd) beats matched fusion (sd), pinball	-0.0009	[-0.0023, +0.0003]	0.384	0.246	not supported
H4-T2	T2: gated (sd) beats matched fusion (sd), pinball	-0.0014	[-0.0026, -0.0003]	0.018	0.012	supported

ID	Statement	Estimate	95% CI	p_{Holm}	p_{BY}	Decision
H4-T3	T3: gated (sd) beats matched fusion (sd), pinball	-0.0025	[-0.0046, -0.0001]	0.033	0.021	supported
H4-T4	T4: gated (sd) beats matched fusion (sd), pinball	-0.0009	[-0.0070, +0.0067]	1.000	1.000	not supported
H4-T5	T5: gated (sd) beats matched fusion (sd), pinball	-0.0478	[-0.0786, -0.0200]	2.1×10^{-6}	1.8×10^{-6}	supported
H5-T1	T1: real context beats null and shuffled context, pinball	-0.0021	[n/a, +0.0021]	0.763	0.573	not supported
H5-T2	T2: real context beats null and shuffled context, pinball	-0.0051	[n/a, -0.0017]	5.4×10^{-4}	4.1×10^{-4}	supported
H5-T5	T5: real context beats null and shuffled context, pinball	-0.0043	[n/a, +0.0285]	1.000	1.000	not supported
H6-T1	T1: online time-adaptive gate lowers scaled |e|	-0.0027	[-0.0068, +0.0020]	0.512	0.353	not supported
H6-T2	T2: online time-adaptive gate lowers scaled |e|	+0.0045	[-0.0003, +0.0100]	1.000	1.000	not supported
H6-T3	T3: online time-adaptive gate lowers scaled |e|	-0.0021	[-0.0062, +0.0023]	0.802	0.668	not supported
H6-T4	T4: online time-adaptive gate lowers scaled |e|	+0.0414	[+0.0007, +0.0949]	1.000	1.000	not supported
H6-T5	T5: online time-adaptive gate lowers scaled |e|	-0.6438	[-0.8178, -0.4410]	5.1×10^{-24}	6.8×10^{-24}	supported
H7-T1	T1: gated model loses less than fusion when RE and weather go missing	-0.0027	[-0.0055, -0.0002]	0.055	0.037	not supported
H7-T2	T2: gated model loses less than fusion when RE and weather go missing	+0.0019	[-0.0005, +0.0045]	1.000	1.000	not supported
H8-T3	T3: carbon block improves pinball loss	+0.0009	[-0.0001, +0.0019]	1.000	1.000	not supported
H8-T1	T1: carbon block does not worsen pinball (margin 1 %)	+0.0009	[-0.0004, +0.0022]	–	–	supported
H8-T2	T2: carbon block does not worsen pinball (margin 1 %)	+0.0006	[-0.0002, +0.0014]	–	–	supported

Table A.3: What publication delay costs: the same gradient-boosted model fitted on the information set with measured delays and on an instant-publication counterfactual.

Target	Horizon	MASE instant	MASE published	Cost (%)	95% CI of the difference	p
T1 energy met	1	0.639	0.825	29.0	[-0.2111, -0.1610]	9.3×10^{-67}
	2	0.829	0.935	12.8	[-0.1186, -0.0940]	3.2×10^{-64}
	3	0.934	1.032	10.4	[-0.1142, -0.0823]	3.8×10^{-33}
	pooled	0.801	0.931	16.2	[-0.1426, -0.1174]	3.3×10^{-107}
T2 actual drawal	1	0.671	0.848	26.4	[-0.1961, -0.1577]	8.4×10^{-85}
	2	0.852	0.966	13.3	[-0.1273, -0.1011]	1.3×10^{-66}
	3	0.966	1.052	8.9	[-0.0976, -0.0756]	1.2×10^{-50}
	pooled	0.830	0.955	15.1	[-0.1369, -0.1142]	7.5×10^{-115}
T3 conventional generation	1	0.485	0.786	62.2	[-0.3258, -0.2775]	6.8×10^{-120}
	2	0.693	0.873	26.0	[-0.1984, -0.1611]	1.3×10^{-76}
	3	0.817	0.934	14.4	[-0.1328, -0.1011]	4.5×10^{-42}
	pooled	0.665	0.865	30.0	[-0.2165, -0.1830]	3.2×10^{-115}
T4 RE generation	1	0.939	1.219	29.8	[-0.3360, -0.2242]	2.3×10^{-24}
	2	1.161	1.302	12.1	[-0.1953, -0.0942]	3.7×10^{-9}
	3	1.259	1.368	8.6	[-0.1584, -0.0660]	4.8×10^{-6}
	pooled	1.120	1.296	15.8	[-0.2249, -0.1331]	2.4×10^{-18}
T5 DAM price	1	0.345	1.164	237.9	[-0.9697, -0.6923]	3.6×10^{-55}
	2	0.474	1.122	136.7	[-0.7958, -0.5254]	1.4×10^{-42}
	3	0.624	1.143	83.1	[-0.6711, -0.3864]	2.9×10^{-19}
	pooled	0.481	1.143	137.7	[-0.8024, -0.5388]	1.7×10^{-44}

Table A.4: Per-seed accuracy of the final forecaster. Every seed is scored against the seasonal naive on the confirmatory period.

Target	Forecaster	seed 0	seed 1	seed 2	seed 3	seed 4	range	seeds better
T1 energy met	stacked weights	0.896	0.899	0.895	0.895	0.895	0.895–0.899	5/5
T2 actual drawal	stacked weights	0.931	0.924	0.926	0.926	0.923	0.923–0.931	5/5
T3 conventional generation	top-k average	0.860	0.860	0.860	0.863	0.861	0.860–0.863	5/5
T4 RE generation	Chronos-2	1.107	1.107	1.107	1.107	1.107	1.107–1.107	5/5
T5 DAM price	top-k average	0.747	0.753	0.743	0.749	0.738	0.738–0.753	5/5

Table A.5: Computing time of the confirmatory run, in minutes; trained models include five seeds and all five targets and three horizons.

Model group	Minutes
LightGBM, direct quantile models	30
XGBoost, multi-quantile	13
BiLSTM	3
TFT	16
PatchTST	6
N-HiTS, TiDE, BiTCN, N-BEATSx	18
Chronos-Bolt Small (zero-shot)	1
Chronos-Bolt Base and Chronos-2 (zero-shot)	69
LA-MCAG and matched fusion, with ablations	76
Combinations and calibration	1
Evaluation and tests	55
All forecasting stages	287

Table A.6: Context-gating comparisons on the confirmatory period, pooled over horizons. A negative difference favours the first model of the pair. Gated: staleness-aware instance gate; fused: matched fixed fusion; matched: trained with staleness-matched source dropout.

Comparison	Target	Metric	Mean A	Mean B	Difference	95% CI	p
Gated (instance) vs fused	T1	mae	0.937	0.934	+0.0030	[-0.0021, +0.0074]	0.939
Gated (instance) vs fused	T1	pinball	0.278	0.279	-0.0009	[-0.0023, +0.0003]	0.038
Gated (instance) vs fused	T2	mae	0.968	0.965	+0.0026	[-0.0034, +0.0075]	0.877
Gated (instance) vs fused	T2	pinball	0.287	0.288	-0.0014	[-0.0026, -0.0003]	0.001
Gated (instance) vs fused	T3	mae	0.870	0.887	-0.0169	[-0.0263, -0.0078]	2.1×10^{-6}
Gated (instance) vs fused	T3	pinball	0.259	0.261	-0.0025	[-0.0046, -0.0001]	0.003
Gated (instance) vs fused	T4	mae	1.226	1.218	+0.0088	[-0.0145, +0.0291]	0.879
Gated (instance) vs fused	T4	pinball	0.382	0.383	-0.0009	[-0.0070, +0.0067]	0.354
Gated (instance) vs fused	T5	mae	1.406	1.700	-0.2951	[-0.3758, -0.2047]	5.2×10^{-28}
Gated (instance) vs fused	T5	pinball	0.249	0.297	-0.0478	[-0.0786, -0.0200]	1.4×10^{-7}
Fused, matched vs round 1	T1	mae	0.934	0.935	-0.0014	[-0.0064, +0.0039]	0.241
Fused, matched vs round 1	T1	pinball	0.279	0.279	+0.0002	[-0.0007, +0.0011]	0.708
Fused, matched vs round 1	T2	mae	0.965	0.966	-0.0007	[-0.0033, +0.0023]	0.273
Fused, matched vs round 1	T2	pinball	0.288	0.287	+0.0014	[+0.0007, +0.0022]	1.000
Fused, matched vs round 1	T3	mae	0.887	0.877	+0.0096	[+0.0046, +0.0154]	1.000
Fused, matched vs round 1	T3	pinball	0.261	0.260	+0.0007	[-0.0003, +0.0017]	0.957
Fused, matched vs round 1	T4	mae	1.218	1.234	-0.0167	[-0.0269, -0.0072]	1.8×10^{-5}
Fused, matched vs round 1	T4	pinball	0.383	0.384	-0.0011	[-0.0038, +0.0015]	0.139
Fused, matched vs round 1	T5	mae	1.700	1.739	-0.0397	[-0.0769, -0.0018]	6.7×10^{-4}
Fused, matched vs round 1	T5	pinball	0.297	0.319	-0.0220	[-0.0312, -0.0118]	5.7×10^{-12}
Gated, matched vs round 1	T1	mae	0.937	0.939	-0.0029	[-0.0058, +0.0001]	0.007
Gated, matched vs round 1	T1	pinball	0.278	0.278	-0.0006	[-0.0013, +0.0001]	0.023
Gated, matched vs round 1	T2	mae	0.968	0.967	+0.0008	[-0.0033, +0.0052]	0.680
Gated, matched vs round 1	T2	pinball	0.287	0.286	+0.0009	[+0.0003, +0.0016]	0.998
Gated, matched vs round 1	T3	mae	0.870	0.865	+0.0044	[+0.0014, +0.0077]	1.000
Gated, matched vs round 1	T3	pinball	0.259	0.259	-0.0004	[-0.0010, +0.0003]	0.071
Gated, matched vs round 1	T4	mae	1.226	1.222	+0.0046	[-0.0065, +0.0144]	0.881
Gated, matched vs round 1	T4	pinball	0.382	0.381	+0.0008	[-0.0022, +0.0048]	0.754

Comparison	Target	Metric	Mean A	Mean B	Difference	95% CI	p
Gated, matched vs round 1	T5	mae	1.406	1.604	-0.2001	[-0.2559, -0.1323]	6.6×10^{-23}
Gated, matched vs round 1	T5	pinball	0.249	0.276	-0.0269	[-0.0412, -0.0105]	9.2×10^{-8}

CRediT authorship contribution statement

Rohit Kumar: Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Investigation, Visualization, Writing – original draft, Project administration, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The State-day panel is built only from reports published by Grid Controller of India, the Central Electricity Authority and the India Meteorological Department. The manifests in the record give the address, retrieval time and SHA-256 hash of every report used, so a copy obtained from the agencies can be checked; some older Central Electricity Authority files are no longer at the addresses recorded in September 2026, because the authority has since moved its website. The processed State-day panel, its data dictionary, the code and the frozen results are openly available on Zenodo (RE-FUSED, version 1.0.0) at `https://doi.org/10.5281/zenodo.23301898` [12], with the

panel and documentation under the CC BY 4.0 licence and the code under the MIT licence. The record holds the code with the frozen results and run logs, the processed datasets, and the larger result files in four archives (parts 1 to 4, each a complete ZIP file that can be unpacked on its own), with SHA-256 checksums of every file.

Code availability and reproducibility

All experiments run from one driver script with fixed seeds and recorded stage logs. The pre-registration, the frozen model choices and the scoring script were hashed and tagged in the code repository before the evaluation data were opened, and every number in this paper is generated from the result files into the text as a macro. The code is released together with the panel; the DOI of the Zenodo record is `https://doi.org/10.5281/zenodo.23301898`. The same code, with the result files, run logs and a folder that lists every file behind this article, is also openly available at `https://github.com/Ohi-Me/RE-FUSED`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the author used Claude (Anthropic) to assist with English language editing, drafting of revised text, consistency checking across tables and narrative text and formatting of the revised manuscript. All experimental design, data collection, model development, the original code implementation, and interpretation of results were

performed entirely by the author. The author carefully reviewed and verified all AI-assisted output and takes full responsibility for the content of this publication.

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